Happiest Health Digital Twin Challenge 2026

TEAM: BLOOM
St. Thomas College

Type 2 Diabetes

DIGITAL TWIN

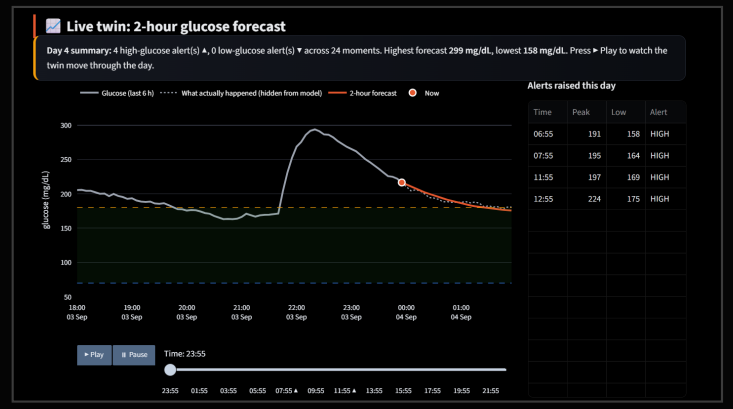
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Everything in this deck comes from our own README and test runs. Where something is untested, we say so.

THE JOURNEY

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Why a doctor needs to see ahead

→

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Fuse sensor and health record in an LSTM

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Risk, results, real people, rules, timeline

WORDS WE USE

Plain meanings, so nothing on the next slides is a mystery

TYPE 2 DIABETES

The body cannot control glucose well, so it swings higher after meals and can drop too low with some medicines.



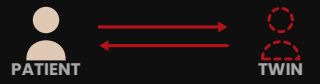
HIGH AND LOW

HIGH means glucose above 180 mg/dL. LOW means below 70 mg/dL. Both are unsafe if they last.

LOW	IN RANGE	HIGH
under 70	70 to 180	over 180

DIGITAL TWIN

A virtual copy of one patient, kept up to date with their data, that we can ask 'what happens next?' and 'what if?'



same data, safe to test

CGM SENSOR

A small wearable that reads glucose every few minutes. Ours reads every 5 minutes, so 72 readings make 6 hours.



1 dot = 1 reading, every 5 min **72 readings = 6 h**

LSTM

A neural network that reads a sequence in order, like the last 6 hours of glucose, and learns the pattern.



reads one step at a time and remembers the earlier ones

SENSOR FUSION

Putting different data together in one model: glucose, steps, heart rate, sleep and the health record.



glucose, steps, heart rate, sleep + record

F1 SCORE

One number that mixes 'how many alerts were right' with 'how many real highs we caught'. 1.0 is perfect.



alerts that were right
real highs we caught
overlap = a good F1

FORECAST ERROR (RMSE)

The typical size of our mistake, in mg/dL. Lower is better.



real glucose
our forecast
gap = error

GLUCOSE

The sugar in your blood. We measure it in mg/dL (milligrams per decilitre).



in your blood
counted in mg/dL

EXECUTIVE SUMMARY

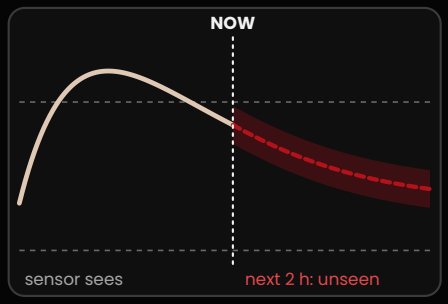
A twin of the patient can warn about a glucose spike before it happens, and show which small change helps most

QUESTIONS

1. How can a doctor know where a patient's glucose will be in the next 2 hours, and not only where it is now?
2. Which small change, such as a short walk or a smaller portion, helps this particular patient the most?
3. How do we show a doctor how sure the forecast is, so it is not trusted blindly?

SITUATION

Most people with Type 2 Diabetes find out about a spike after it has happened. A sensor shows the glucose right now. It does not show the next two hours, and that is the window when a walk or a smaller meal can still change the result. A missed low is even more dangerous than a high.



SOLUTION

A **fusion LSTM** reads 6 hours of sensor data and the patient's health record together, and forecasts the next 2 hours.

Alerts for **HIGH (above 180)** and **LOW (below 70)**, with an extra safety margin for lows and an uncertainty band.

A **what-if engine** re-runs the forecast with one change: a 10-minute walk, a 30% smaller portion or a low-GI swap.

- FORECAST 2 H
- HIGH / LOW ALERT
- WHAT-IF REMEDY

IMPACT

- The doctor sees a rise or a low before it happens.
- Every screen says what the model is sure about and what it is not.
- Remedies are shown as model predictions, with no doses calculated. The doctor decides.

2 h	6 h	10
FORECAST WINDOW	OF HISTORY READ	SIGNALS FUSED

Research prototype trained on synthetic data. Not medical advice.

STAKEHOLDERS' ANALYSIS

101 M

PEOPLE IN INDIA LIVING WITH DIABETES



1 dot = about 1 million people

A large national study (ICMR-INDIAB, 2023) also found about 136 million with prediabetes. Most are managed by a doctor who sees them for a short visit every few months, with only a few readings to look at.

70%+

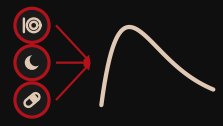
OF THE DAY SHOULD BE IN RANGE



International guidance on continuous glucose monitoring asks for at least 70% of the day between 70 and 180 mg/dL, and less than 4% of the day below 70. Our dashboard shows exactly these numbers for every patient.

2 in 1

FOOD, SLEEP AND MEDICINE TOGETHER



Glucose depends on food, activity, sleep, stress and medicines all at once. Two patients with the same curve can need different advice, so the model needs the health record as well as the sensor.

PRIMARY RESEARCH

Conversation with a hospital nurse

We asked a nurse who cares for diabetic patients what slows her down. She has to keep checking each patient's blood sugar, read the doctor's notes every time, and look up which medicines the patient takes. She also wants one place that shows which patients need which insulin shot, and how strong, before or after meals. Her other problem is paper. She handles files and reports all day, and sometimes a record gets torn or damaged.

- Check blood sugar every time
- Read the doctor's notes
- Which medicine, which insulin shot, how strong
- Paper files that get torn

MAJOR PAIN POINTS

1

A doctor sees what already happened, not what is about to happen.

2

A missed low is the more dangerous mistake, so alerts must lean towards catching lows.

3

A forecast with no sign of how sure it is can be trusted too much.

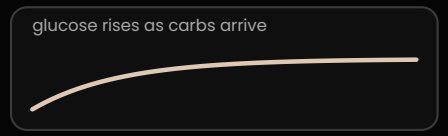
4

Advice must be reference only, because the doctor knows the patient and the model does not.

WHAT MOVES GLUCOSE DURING THE DAY

Glucose is the sugar in your blood. It goes up and down all day. Here are the five things that push it

1 MEAL



WHAT HAPPENS
Food has carbs (sugars and starches like rice, roti, sweets). The body breaks them into glucose, which goes into the blood.

WHY IT MATTERS
Some foods raise glucose fast, some slowly. This speed is the glycaemic index. Example: white rice and mithai are fast. Dal and vegetables are slower.

OUR TWIN
Knows the carbs and GI of each meal (Indian menu).

2 AFTER THE MEAL



WHAT HAPPENS
Glucose does not jump at once. It climbs slowly, peaks about 1 to 2 hours after eating, then comes down.

WHY IT MATTERS
In Type 2 Diabetes the body cannot bring it down well, so the peak can cross 180 mg/dL. That is HIGH.

OUR TWIN
Watches exactly this window. It looks 2 hours ahead and warns the doctor before the peak.

3 WALK



WHAT HAPPENS
When you walk, muscles use glucose as fuel, so glucose in the blood goes down.

WHY IT MATTERS
Even a 10-minute walk after eating helps. A walk right after the meal helps more than one much later, so timing matters.

OUR TWIN
Reads steps and heart rate, plus 3 walk-timing features. The doctor can test 'what if a 10-minute walk?' and see the new curve.

4 SLEEP

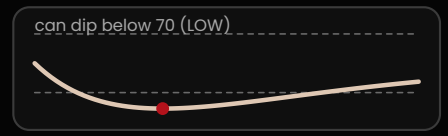


WHAT HAPPENS
After a bad night the body reacts worse to sugar. Early morning hours have their own pattern too, as the body releases extra sugar-raising hormones.

WHY IT MATTERS
The same breakfast can raise glucose more after a bad night than after a good night.

OUR TWIN
Sleep and heart rate variability (HRV, how steady the heartbeat is) are two of its 10 signals.

5 MEDICINE



WHAT HAPPENS
Metformin helps the body use sugar. Insulin and sulfonylureas push glucose down strongly.

WHY IT MATTERS
They can push it too low (LOW, below 70 mg/dL), and a low can be dangerous.

OUR TWIN
Reads the medicine list from the health record. The doctor can edit it. A sensor alone can never tell this.

SIGNALS USED IN EACH MOMENT

1 MEAL	2 AFTER THE MEAL	3 WALK	4 SLEEP	5 MEDICINE
Carbs, glycaemic index, glucose	Glucose, heart rate, steps	Steps, heart rate, walk timing	Sleep, HRV, glucose	Medicine list from the health record

Research prototype trained on synthetic data. Not medical advice.

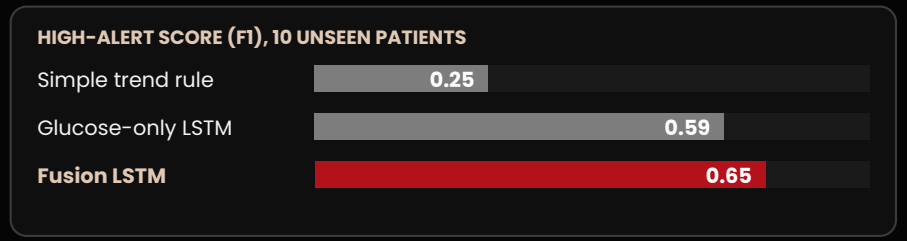
THE BIG IDEA

Each data source tells part of the story. Fusing the sensor stream with the health record gives a forecast that no single source can give

What we need	Glucose sensor	Steps and heart rate	Sleep and HRV	Health record	After fusion
Glucose now	🟢	🔴	🔴	🔴	🟢
Trend over the last hours	🟢	🟡	🟡	🔴	🟢
Knows the patient's usual behaviour	🟡	🟡	🟡	🟢	🟢
Knows the medicines	🔴	🔴	🔴	🟢	🟢
Explains why glucose moved	🔴	🟢	🟡	🟡	🟢
Forecast of the next 2 hours	🟡	🔴	🔴	🔴	🟢

Green: does this well. Yellow: partly. Red: not at all. This is our own reasoning, not a measured result.

Fusion fills in the gaps of each source. For example, the glucose sensor alone can show a rise, but only the health record tells the model that this patient is on insulin and may drop low. Putting both into one model gives a more complete picture. We also tested this honestly: on the synthetic test patients, fusion did not reduce the 2-hour forecast error (18.57 vs 18.42 mg/dL), but it raised the HIGH-alert score from 0.59 to 0.65.



UNIQUE SELLING POINT OF THE PRODUCT: the model-suggestive remedy is the most unique and useful part of the project. It helps the doctor reach the best decision and give better advice and suggestions to diabetic patients.

WHY AN LSTM

We tested the LSTM against simple baselines. We did not test the other models, and we say so

LSTM

OUR CHOICE



A neural network that reads a sequence in order. It fits a time series like glucose, runs on an ordinary CPU, is a standard choice for glucose forecasting, and lets us combine the sensor stream with the patient record in one model.

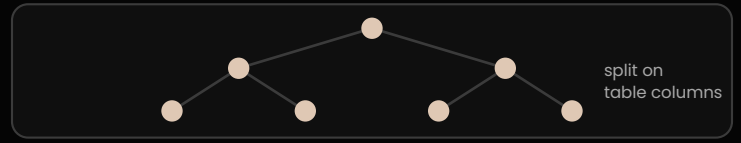
OUR READING, NOT A MEASURED RESULT

- Reads a sequence in order
- Works with modest data
- Light on compute
- Fits sensor + record together

TESTED BY US | COMPUTE: LOW

TREE MODELS

SUCH AS XGBOOST



Fast and good with tables, but they do not read a sequence in order, so we would have to build the time features by hand.

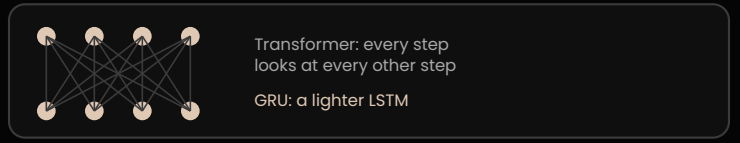
OUR READING, NOT A MEASURED RESULT

- Reads a sequence in order
- Works with modest data
- Light on compute
- Fits sensor + record together

NOT TESTED BY US | COMPUTE: LOW

GRU AND TRANSFORMER

SEQUENCE MODELS



Also read sequences. A GRU is a lighter LSTM. A Transformer usually needs a lot more data than we have.

OUR READING, NOT A MEASURED RESULT

- Reads a sequence in order
- Works with modest data
- Light on compute
- Fits sensor + record together

NOT TESTED BY US | COMPUTE: MEDIUM TO HIGH

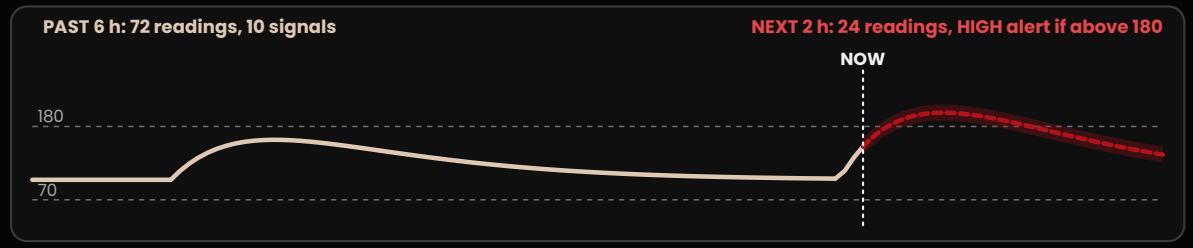
Honest note. We compared the LSTM with two simple baselines (no change, and a trend line). A comparison with a GRU or a Transformer is listed as a next step.

HOW OUR IDEA WILL WORK

Our idea has two steps: 1. Forecast and alert. 2. Test a small change before recommending it

1 FORECAST AND ALERT

The fusion LSTM reads the last **6 hours (72 readings)** of **10 sensor signals**, the patient's health record and 3 walk features. It forecasts the next **2 hours (24 readings)** as a change from the current glucose, which is easier for the network to learn. A **HIGH alert** is raised when the forecast crosses 180 mg/dL. A **LOW alert** is raised when it comes close to 70 mg/dL.

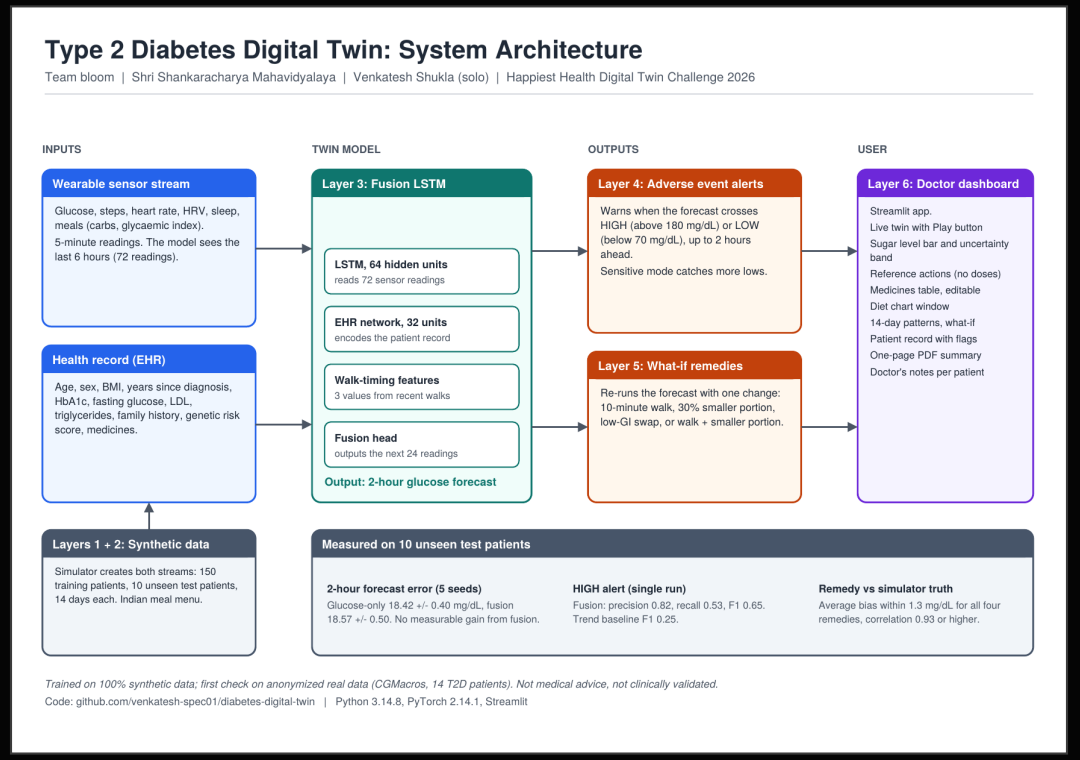


2 MODEL SUGGESTIVE REMEDIES

The model re-runs the forecast with **one change**: a 10-minute walk, a 30% smaller portion, a low-GI swap, or a walk plus a smaller portion. The doctor sees the curves side by side. We checked the answers against the simulator's true answer on unseen patients. The average error was within about **1.3 mg/dL**.

- 10-minute walk
- 30% smaller portion
- Low-GI swap (GI 50)
- Walk + smaller portion

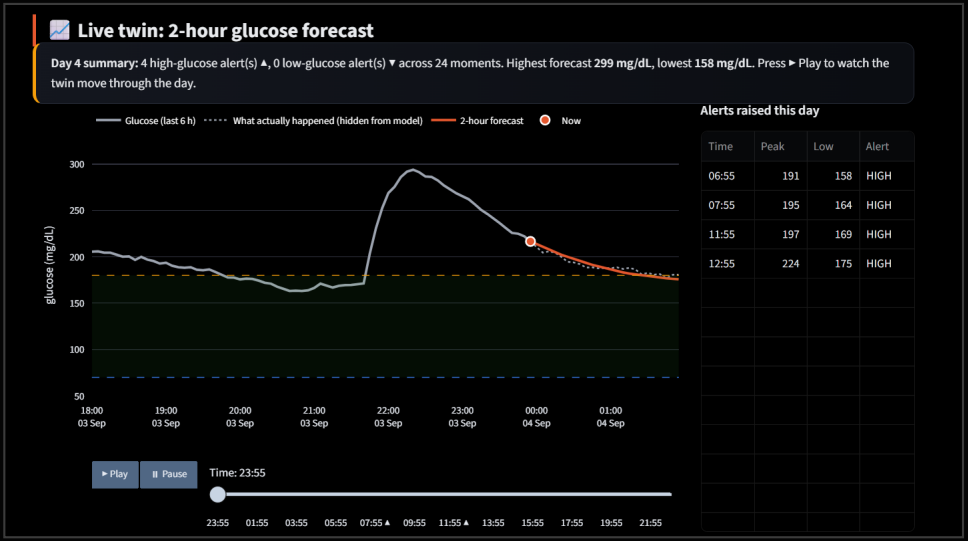
ARCHITECTURE WE USED



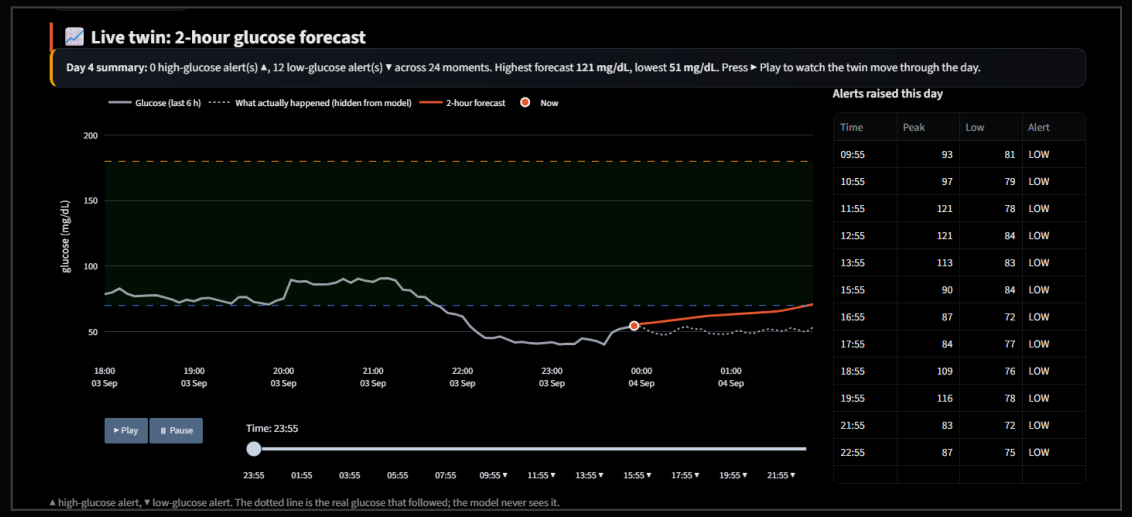
The model's job is to support the doctor. Nothing in the dashboard is a prescription.

THE DOCTOR'S DASHBOARD: LIVE TWIN

Play the day and watch the forecast appear 2 hours ahead of the real curve



HIGH ALERT (T03, DAY 4: 4 ALERTS)



LOW ALERT (T06, DAY 4: 12 ALERTS)

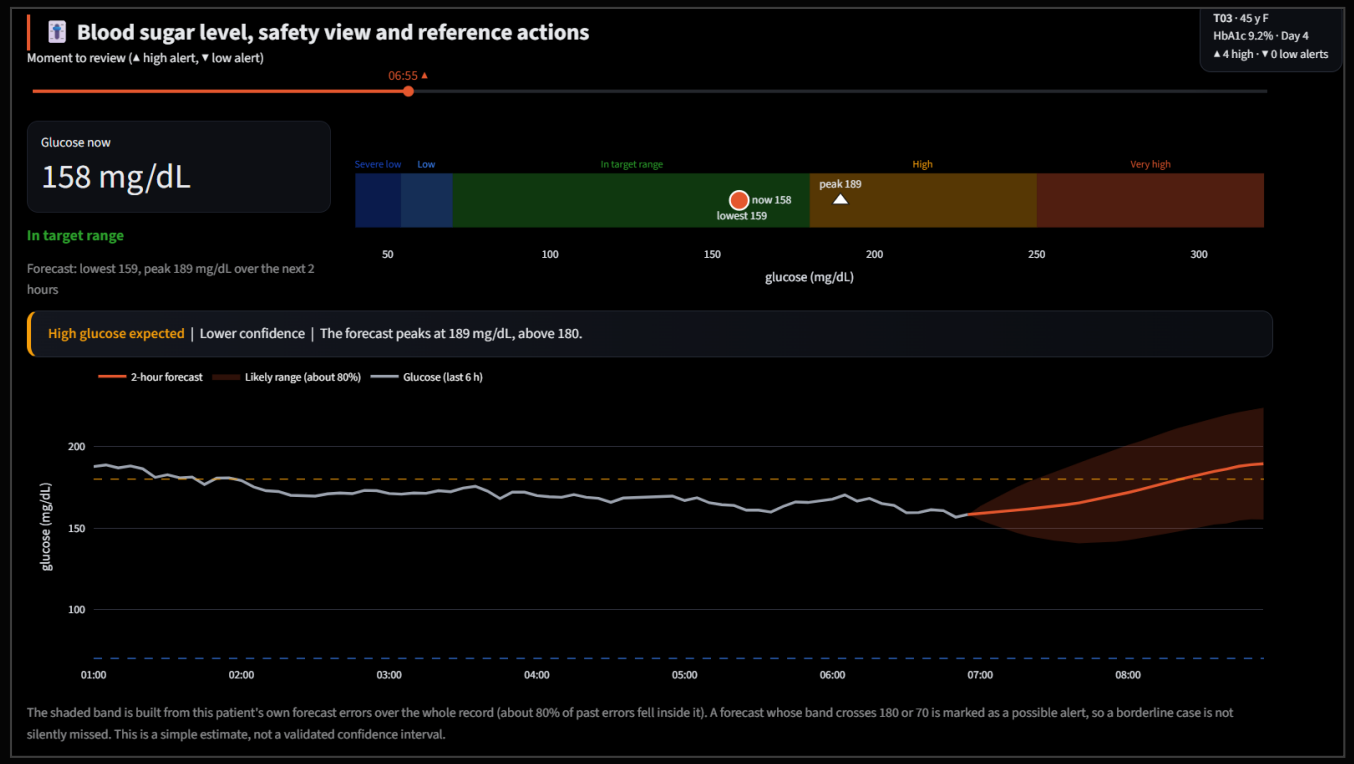
The play button replays the day. The forecast appears 2 hours ahead of the real curve, so the doctor can see the warning arrive before the event.

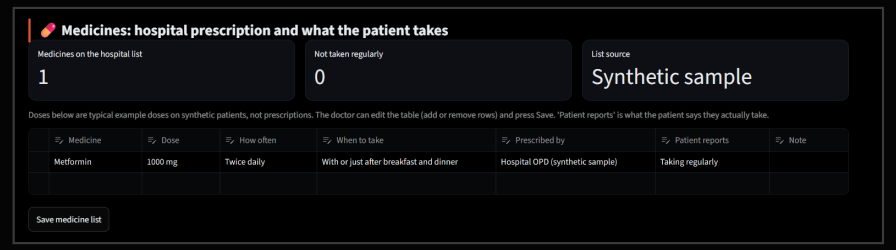
Sensitive mode is the default, because a missed low is the more dangerous error. It warns 15 mg/dL before the 70 line.

Every alert is a model prediction for the doctor to read, not an instruction.

SAFETY VIEW AND MEDICINES

Where glucose is now, where it is heading, and what the patient is taking





- A coloured bar shows where glucose is now and the forecast low and peak.
- The shaded band is the likely range (about 80%). A wide band means lower confidence, and it is flagged.
- The doctor edits the medicine list. The patient's own report sits next to it.
- Reference actions only. No doses are calculated.

MODEL SUGGESTIVE REMEDY AND 14-DAY PATTERNS

Pick a meal, compare the remedies side by side, then see how the whole fortnight looked





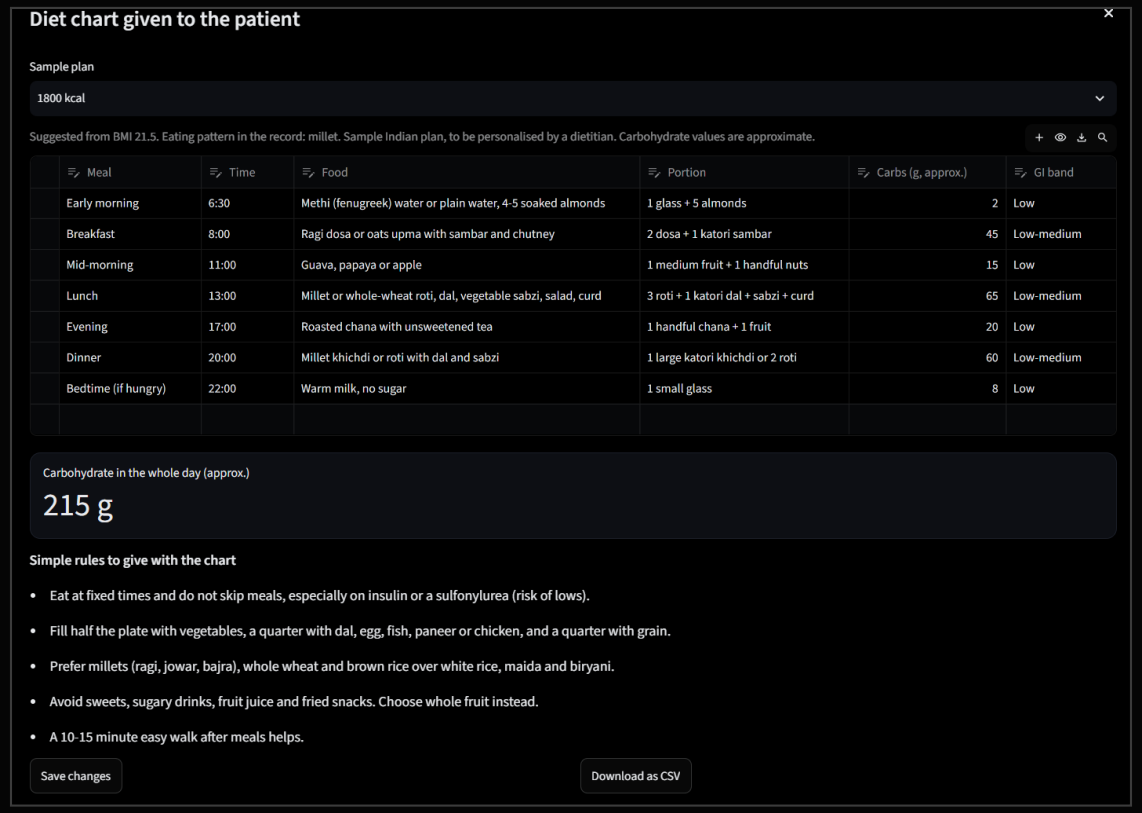
T03 is shown with a 138 g-carb meal, an extreme case. On average, a walk plus a smaller portion lowers the peak by about **20 mg/dL**.

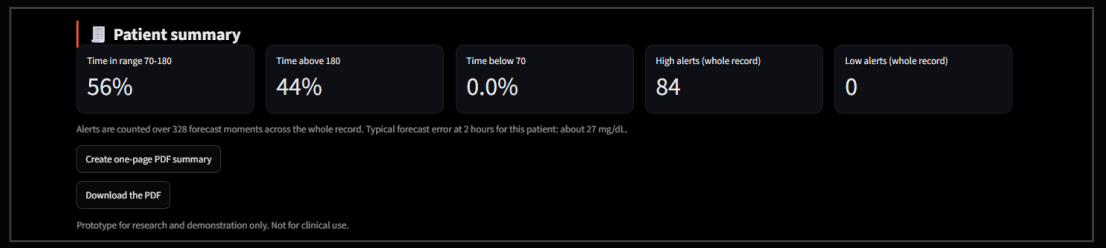


T03 over 14 days: **56%** of the time in range, **44%** above 180.

DIET CHART AND PATIENT SUMMARY

The doctor's hand-over: a diet plan to edit and a one-page summary to print





- The diet chart opens in its own window. It is a sample Indian plan; the doctor edits it and saves or downloads it.
- The summary shows time in range, time above 180, time below 70 and alert counts.
- One click makes a one-page PDF to hand over.
- Marked as a prototype, not for clinical use.

HOW AN ALERT IS DECIDED

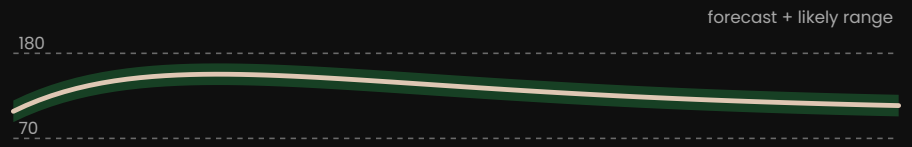
Three levels make sure a borderline case is not silently missed, and the doctor can always see how sure the model is

SENSOR STREAM → HEALTH RECORD → FUSION LSTM FORECAST → UNCERTAINTY BAND → ALERT LEVEL

LEVEL 1

NORMAL

The forecast and its likely range (about 80%) stay between 70 and 180 mg/dL. No alert.



LEVEL 2

POSSIBLE ALERT, LOWER CONFIDENCE

The forecast stays inside 70 to 180 but the likely range crosses a line. The dashboard marks it as a possible alert, so a borderline case is not hidden.



LEVEL 3

ALERT

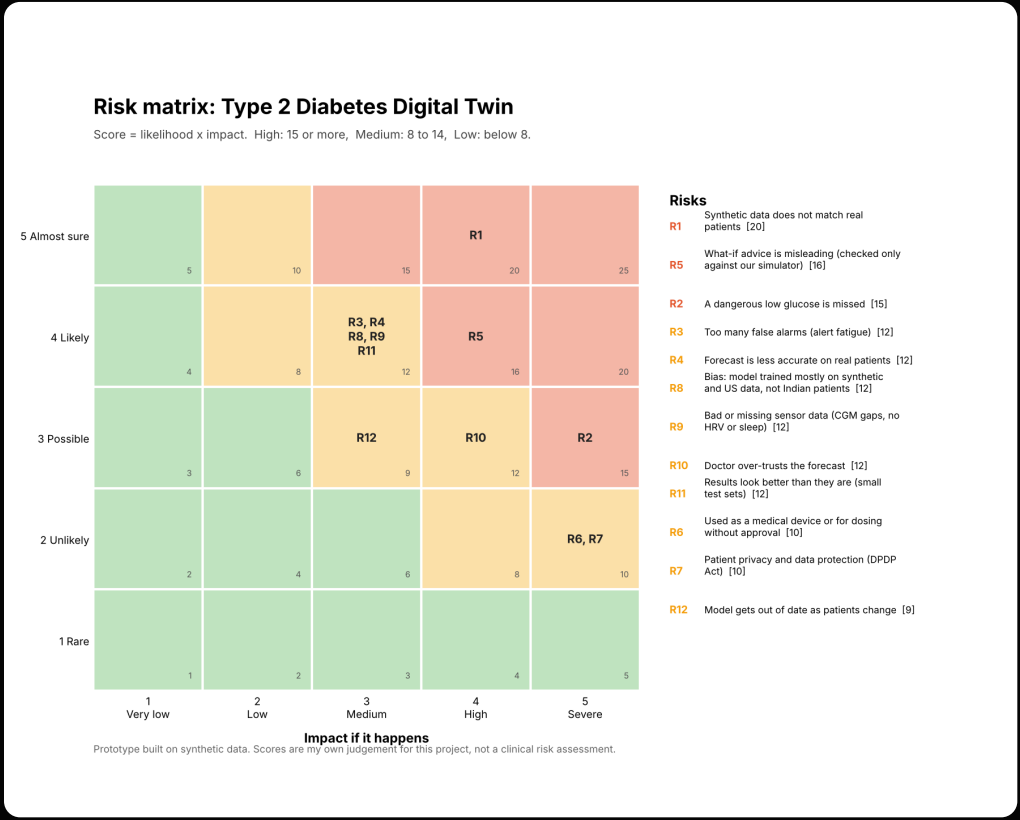
The forecast itself crosses 180 (HIGH) or comes within 15 mg/dL of 70 (LOW, sensitive mode). The dashboard shows the reference actions for the doctor.



⚠ **Why 15 mg/dL?** For lows we chose to warn earlier, because a missed low is the more dangerous mistake. The cost is more false alarms.

PRIORITISATION OF RISK

12 risks scored as likelihood x impact, each from 1 to 5. Red is 15 or more



Risk	L	I	Score	What we did
Synthetic data does not match real patients	5	4	20	Tested on real CGMacros data. Next step: Indian patient data
What-if advice is only checked against our simulator	4	4	16	Labelled as model predictions, no doses
A dangerous low is missed	3	5	15	Sensitive low mode, uncertainty band, doctor in the loop

The scores are our own judgement for this project. They are not a clinical risk assessment.

TOP THREE SCORES



SYSTEM FLOW

Spot the risk → Reduce it in the design → Show the doctor what is still uncertain

RESULTS ON 10 UNSEEN PATIENTS

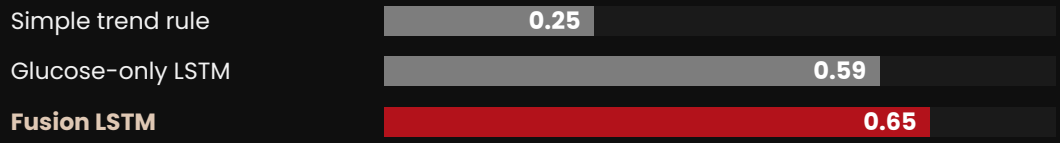
Where the model helps, and where it does not

2-HOUR FORECAST ERROR (5 REPEATS), LOWER IS BETTER



Fusion gives **no measurable gain** in forecast error, and we say so.

HIGH ALERTS (F1 SCORE), HIGHER IS BETTER



82% of the HIGH alerts it raises are correct. It catches **53%** of highs in default mode and **85%** in sensitive mode.

LOW ALERTS: SHARE OF LOWS CAUGHT



Sensitive mode gives more false alarms. That is why the dashboard uses it.

WHAT-IF REMEDIES: CHECK AGAINST THE SIMULATOR

1.3

MG/DL AVERAGE ERROR

within 1.3 mg/dL

CORRELATION



Checks use the same simulator that made the data.

Results and accuracy depend on the quality and quantity of the data. This project is trained on synthetic data, so it can show lower accuracy on real patients.

Alert numbers come from a single run. Remedy checks use the same simulator that made the data, so they show consistency, not real physiology.

WHAT HAPPENS ON REAL PEOPLE

We ran the trained model, with no retraining, on CGMacros: anonymised real data from 45 adults on PhysioNet

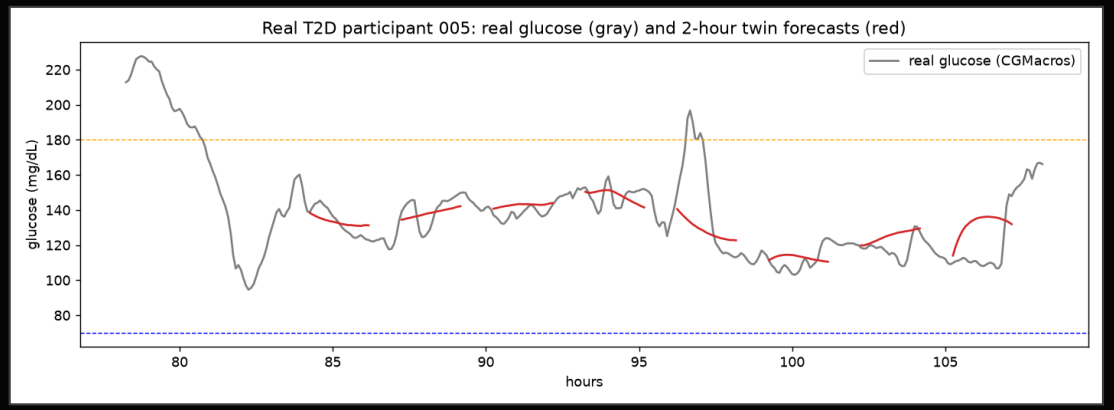
2-HOUR ERROR (MG/DL), 14 PEOPLE WITH TYPE 2 DIABETES

Assume glucose stays the same	52.2
Trend line	107.0
Sensor-only LSTM	46.1
Our fusion twin	46.0

18 > 46
synthetic vs real error in mg/dL, about 2.5 times larger

12%
better than 'no change' on real people

8% / 37%
of real highs caught, default / sensitive mode



On synthetic patients the 2-hour error is about 18 mg/dL. On real people it is 46. It is still about **12% better** than guessing that glucose stays the same, so the model learned real glucose behaviour. After fine-tuning on real people and testing on people it had never seen, the error dropped by about **4 to 9%**. That is our honest gap between synthetic and real data, and closing it is the next step.

Accuracy depends on the quality and quantity of the training data. We trained on synthetic data, so expect lower accuracy on real patients, as the numbers above show.

REGULATION WE RESEARCHED

What it would take to use this in India

Software that monitors or forecasts glucose for a doctor counts as **Software as a Medical Device** under the Medical Devices Rules, 2017. **CDSCO**, India's regulator, published its medical device software guidance in 2026.

The class, from A to D, depends on how serious the condition is and how much the software influences the doctor. Our tool only informs the doctor, so our reading is **Class A or B**, and CDSCO makes the final call.

DEVICE CLASS: LOWER TO HIGHER RISK



Class A and B: apply on Form MD-3, licence MD-5 from the State authority.

Class C and D: apply on MD-7, licence MD-9 from the Central authority.

Rule	What it asks		What we did
ISO 13485	A documented quality system	🟡	README with build order, pinned versions, full code history
IEC 62304	A controlled software lifecycle	🟡	One job per script, fixed seeds, 5-seed test, reproducible models
ISO 14971	A risk management file	🟡	12-risk matrix, a first draft
DPDP Act	Consent, minimum data, security	🟢	Synthetic and anonymised open data only
ABDM / ABHA	Link to hospital health records	🔴	Not built, a later step

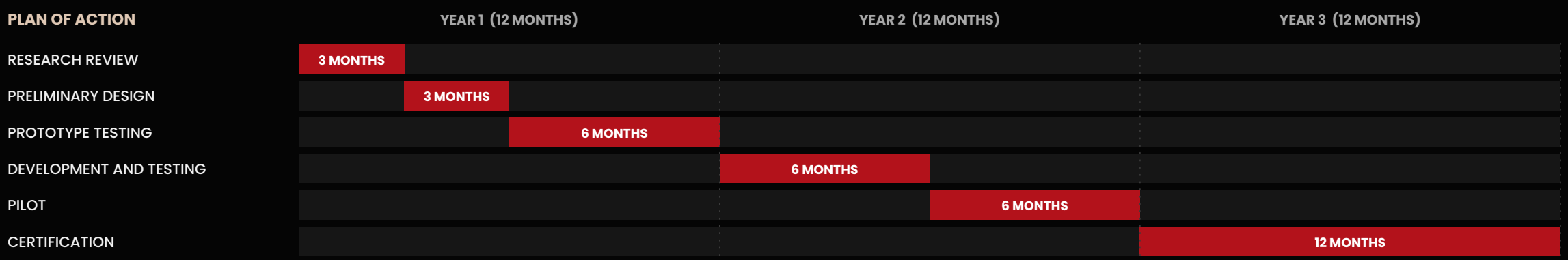
Green: followed in the prototype. Yellow: partly followed. Red: not built. Our own reading.

⚠ **Today this is a research prototype.** We have applied for no licence and have not been audited against any of these standards.

EXECUTION STRATEGY AND TIMELINE

This is our proposed plan, not something already done

1	**Research review**	Read the literature on glucose forecasting and on hospital use in India. Find Indian CGM datasets and a clinical partner.
2	**Preliminary design**	Write the full risk file (ISO 14971) from our 12-risk matrix. Agree what the doctor needs to see.
3	**Prototype testing**	Retrain and test on real Indian patient data. Compare against a GRU and a Transformer. Add prediction intervals and a Clarke Error Grid.
4	**Development**	Build the software under IEC 62304. Tune the alert thresholds on a separate validation set. Count alerts per episode and the minutes of warning.
5	**Pilot**	Silent pilot in a hospital with doctors, only with ethics approval. The doctor sees the forecast but nobody acts on it yet.
6	**Certification**	Prepare the quality system (ISO 13485) and apply to CDSCO.



THANK YOU.

github.com/venkatesh-spec01/diabetes-digital-twin

REFERENCES

		REFERENCE	WHAT WE USED IT FOR
01	STUDY	Anjana, R. M. et al., 2023, "Metabolic non-communicable disease health report of India: the ICMR-INDIAB national cross-sectional study", The Lancet Diabetes & Endocrinology.	101 M people with diabetes, 136 M with prediabetes
02	GUIDELINE	Battelino, T. et al., 2019, "Clinical targets for continuous glucose monitoring data interpretation: recommendations from the international consensus on Time in Range", Diabetes Care.	70% of the day in range, under 4% below 70
03	DATASET	Gutierrez-Osuna, R., Kerr, D., Mortazavi, B. and Das, A., 2025, "CGMacros: a scientific dataset for personalized nutrition and diet monitoring" (version 1.0.0), PhysioNet. https://doi.org/10.13026/3z8q-x658	The test on real people (14 with Type 2)
04	PAPER	Hochreiter, S. and Schmidhuber, J., 1997, "Long Short-Term Memory", Neural Computation.	The LSTM model
05	LAW	Ministry of Health and Family Welfare, India, Medical Devices Rules, 2017.	Device classes A to D, forms and licences
06	REGULATOR	CDSCO, Guidance Document on Medical Device Software, 2026.	Rules for medical software
07	STANDARD	IEC 62304, ISO 14971 and ISO 13485 (international standards).	Lifecycle, risk and quality system
08	LAW	Digital Personal Data Protection Act, 2023, India.	Consent and data protection